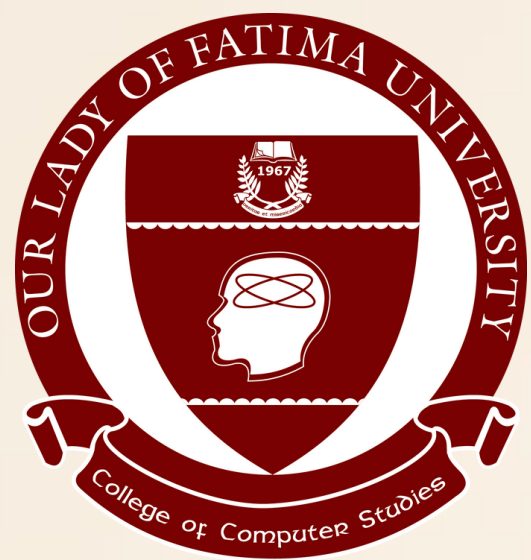




Module 8

Internet and World Wide Web

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



The Internet and World Wide Web (WWW) are two of the most important innovations that connect people, information, and services worldwide. While the Internet is the global network that links millions of computers, the World Wide Web is a system of interlinked documents and resources accessed through the Internet.

MODULE 8

In this module, you will learn:

- 1. The difference between the Internet and the World Wide Web.**
- 2. How domain names and URLs help us locate websites.**
- 3. The process of how the web works using the client-server model with HTTP/HTTPS.**
- 4. The role of web browsers and web servers in accessing and delivering online content.**

What is Internet?



The Internet is a global network of interconnected computers and devices that allows users to access and share information and devices, allowing them to communicate and exchange data.



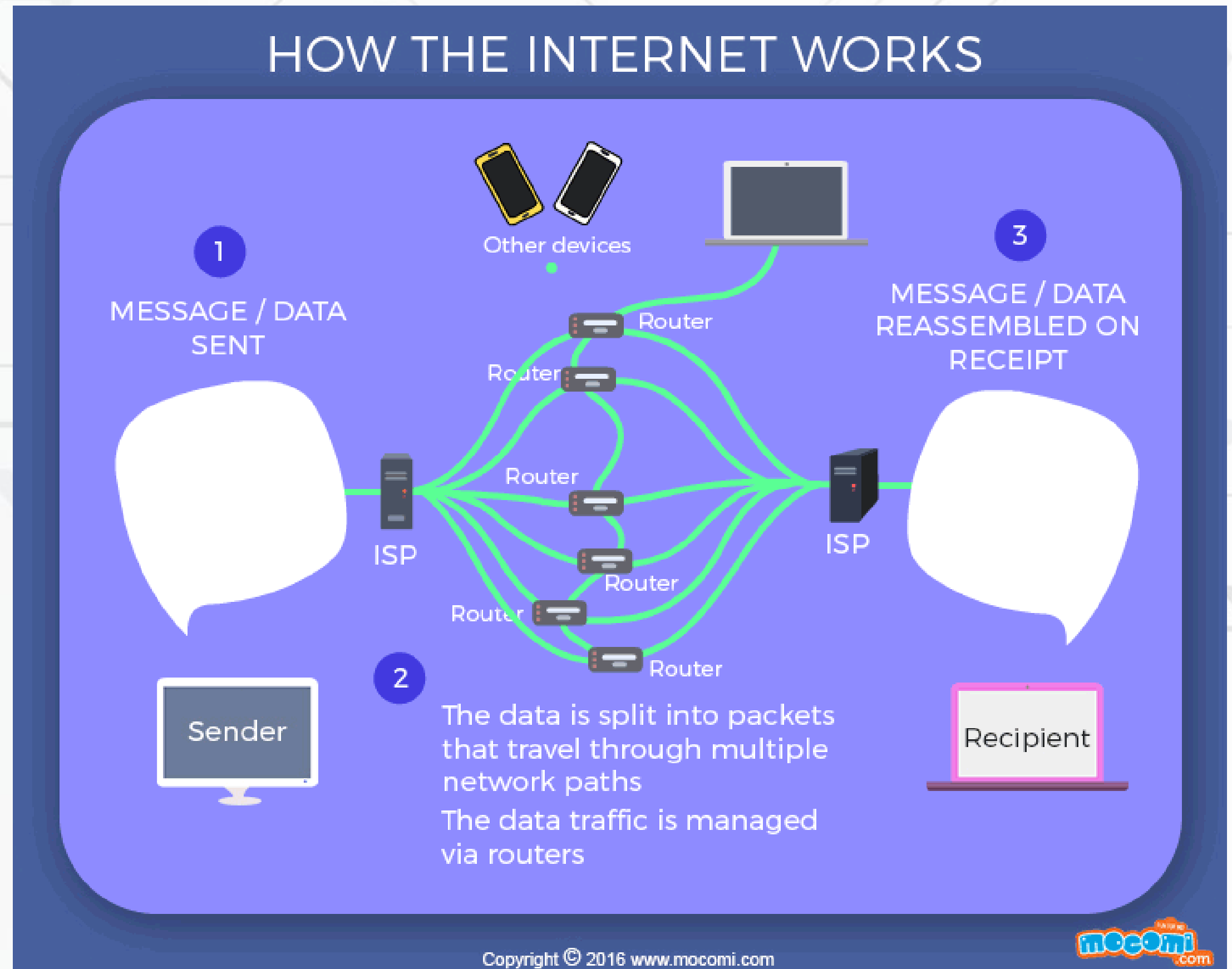
History of Internet



The Internet came in the year 1960 with the creation of the first working model called **ARPANET (Advanced Research Projects Agency)**. It allowed multiple computers to work on a single network which was their biggest achievement at that time.

ARPANET uses packet switching to communicate multiple computer systems under a single network. In October 1969, using ARPANET first message was transferred from one computer to another. After that technology continues to grow.

How Does the Internet Work?



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World Wide Web (WWW)



The World Wide Web is a collection of web pages and documents accessed through URLs (Uniform Resource Locators) on the Internet. For example, www.google.com is a URL, and its content is part of the WWW. It serves as an information retrieval system where documents are linked using hypertext or hypermedia. Hyperlinks allow users to click on words or phrases to easily access related information.

The World Wide Web was created by Timothy Berners-Lee in 1989 at CERN to help researchers collaborate efficiently. The World Wide Web Consortium (W3C) was later established for its continued development.

Difference Between World Wide Web and the Internet

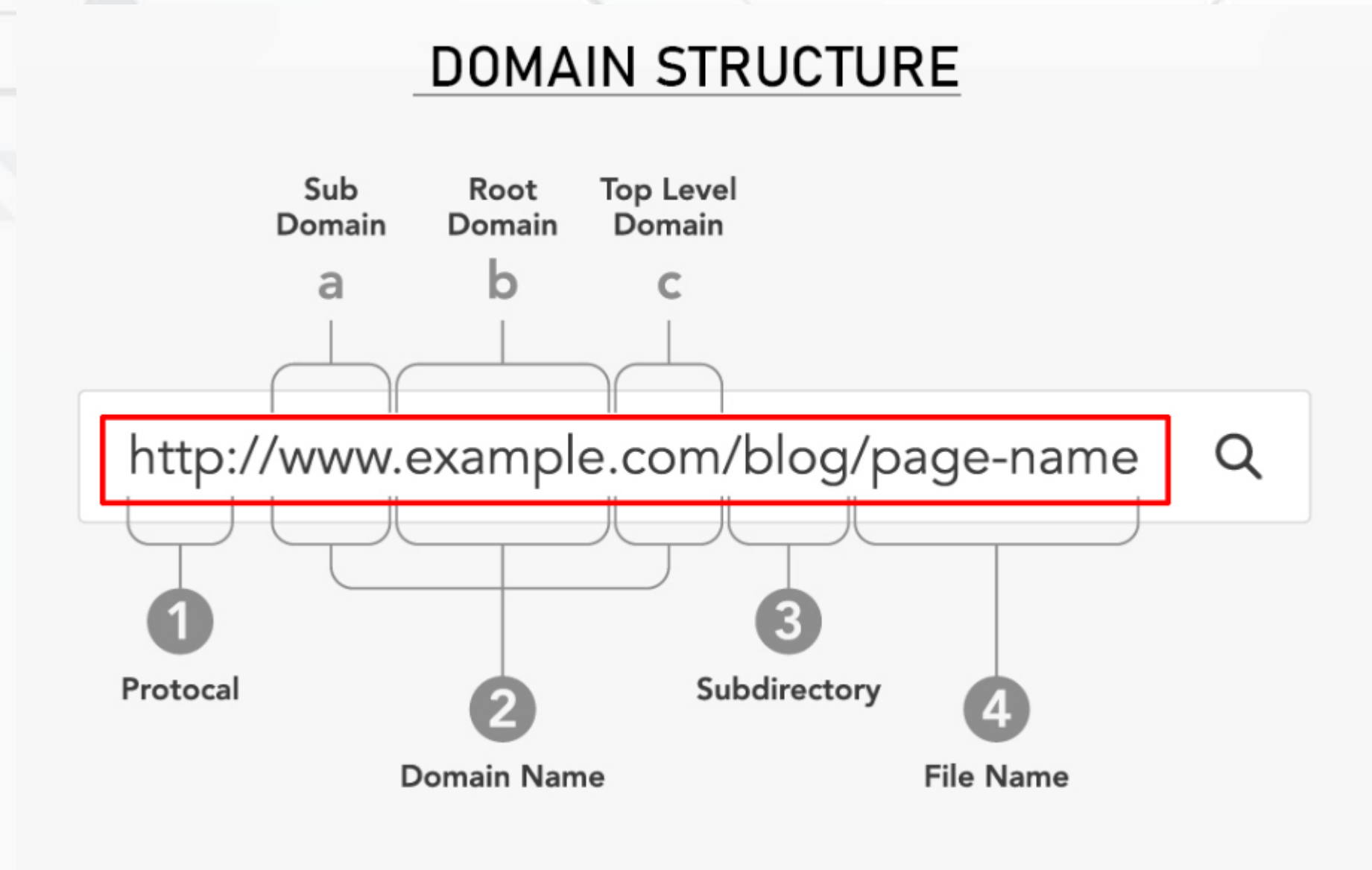


WORLD WIDE WEB	INTERNET
All the web pages and web documents are stored there on the World wide web and to find all that stuff you will have a specific URL for each website.	The Internet is a global network of computers that is accessed by the World wide web.
The world wide web is a service.	The Internet is an infrastructure.
The world wide web is a subset of the Internet.	The Internet is the superset of the world wide web.
The world wide web is software-oriented.	The Internet is hardware-oriented.
The world wide web uses HTTP .	The Internet uses IP Addresses .
The world wide web can be considered as a book from the different topics inside a Library.	The Internet can be considered a Library.
Examples: Websites, e-commerce, blogs	Example: The network connecting all online services

What is a Domain Name?



It is very hard to remember the IP address of the server. Hence, the domain name is the text form of the IP address and it is more human friendly and easy to learn. It is a part of the URL and it is mostly used for branding the organization. All the parts of the domain name are separated by periods or dots.



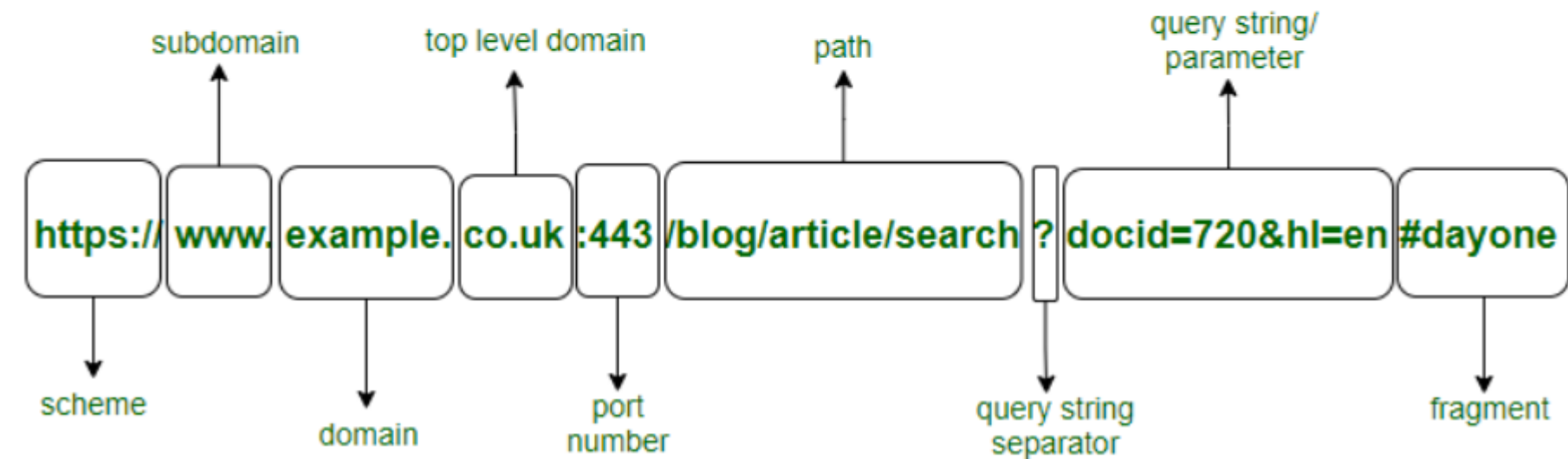
What is Uniform Resource Locator? (URL)



Uniform Resource Locator is known as a URL. It is a string that provides the complete address of the web page i.e, this is also known as a web address. In order to access a particular website you just need to put the URL in the search bar to find the website.

Parts of a URL

URL : <https://www.example.co.uk:443/blog/article/search?docid=720&hl=en#dayone>



Difference Between Domain Name and URL

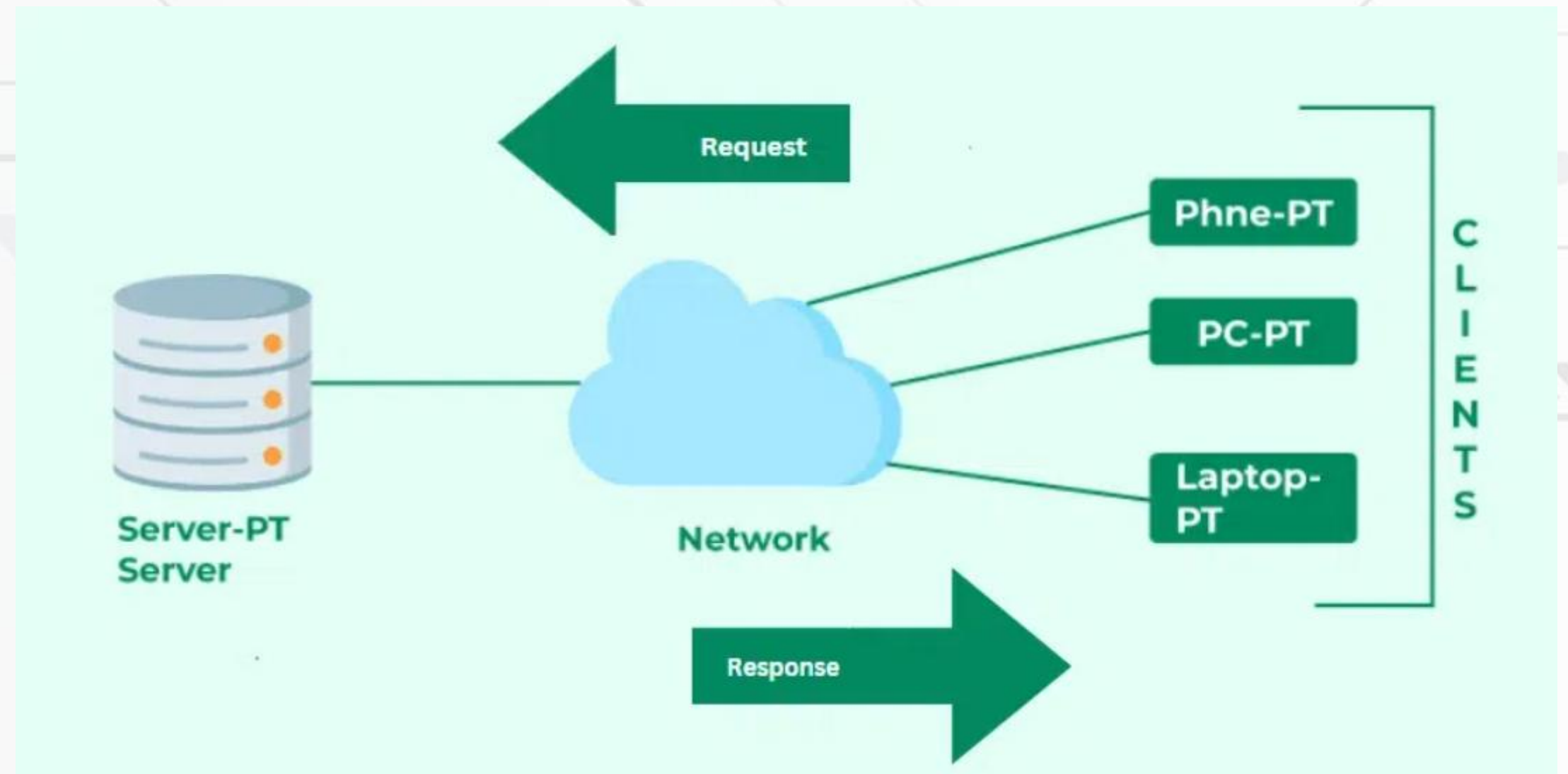


Basis of Comparison	Domain Name	Uniform Resource Locator
Basics	The main motive of the domain name is to make it easier to access a website. They are more used as a brand and are usually referred to as a brand name.	If you want to find a particular website URL is best for it because a URL is a complete web address to find a particular website.
Address	A domain name is a human-friendly text form of the IP address.	URL is a string that represents the complete web address of any web page.
Represented as	It is the part of the URL that is more human friendly.	It is the string that represents a complete web address that contains the domain name.
Consists of	It contains three parts-Top Level Domain, Intermediate Level, and the Low Level.	It also contains the following parts-method, protocol, hostname, port, and path of the file.
Example	Example: geeksforgeeks.org	Example: https://www.geeksforgeeks.org/
Year	The first domain name registration was Symbolics.com by on 15th March, 1985.	URL was defined by Tim Berners Lee in the year 1994.
Working	A domain name is used in conjunction with a domain name system, which includes the top-level domain, DNS caching, and an authoritative DNS server.	URLs, on the other hand, are made up of a variety of elements such as protocols, parameters, domain names, path ports, and so on.

Client-Server Model



The Client-Server Model is a distributed architecture where clients request services and servers provide them. It underpins many modern systems, including websites, email, and cloud storage platforms.



How Does the Client-Server Model Work?



Client: A client is any device or software that initiates communication by requesting data or services from a server. Common client applications include:

- Web browsers (e.g., Chrome, Firefox)
- Email apps (e.g., Gmail, Outlook)

Server: A server is a powerful systems that listens for and responds to client requests by delivering data or performing tasks. Servers often handle multiple simultaneous client requests. Common server applications include:

- Web Servers (e.g., Apache, Nginx)
- Email Servers
- Database servers

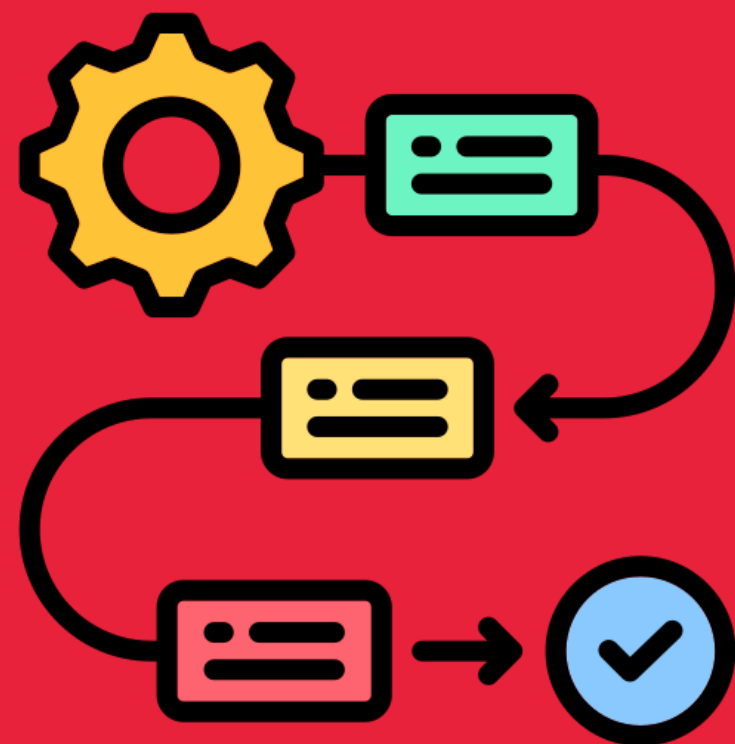
Hypertext Transfer Protocol (HTTP)



HTTP (Hypertext Transfer Protocol) is a fundamental protocol of the Internet, enabling the transfer of data between a client and a server. It is the foundation of data communication for the World Wide Web.

HTTP provides a standard between a web browser and a web server to establish communication. It is a set of rules for transferring data from one computer to another. Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP. It is an application protocol that is used for distributed, collaborative, hypermedia information systems.

Methods of HTTP



GET

Used to retrieve data from a specified resource.

POST

Used to submit data to be processed by a specified resource.

PUT

Used to update or create a resource on the server.

PATCH

Similar to PUT but used for partial modifications to a resource.

DELETE

Used to remove a specified resource from the server.

HEAD

Similar to GET but retrieves only the response headers, useful for checking resource properties without transferring the full content.

OPTIONS

Used to retrieve the communication options available for a resource, including supported methods and headers.

TRACE

Used for debugging purposes to echo the received request back to the client, though it's rarely used due to security concerns.

CONNECT

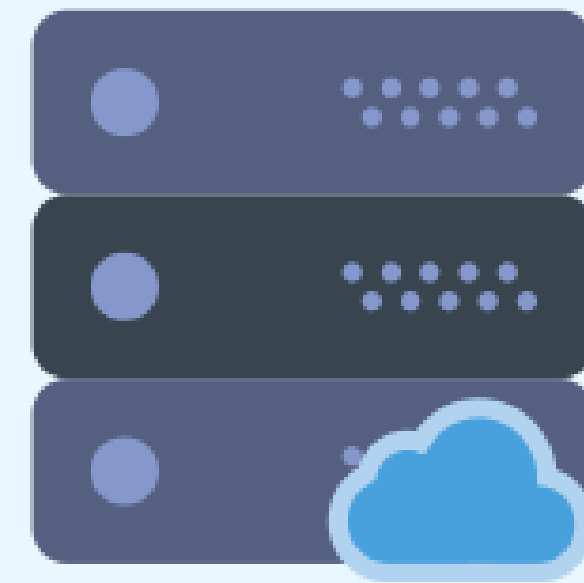
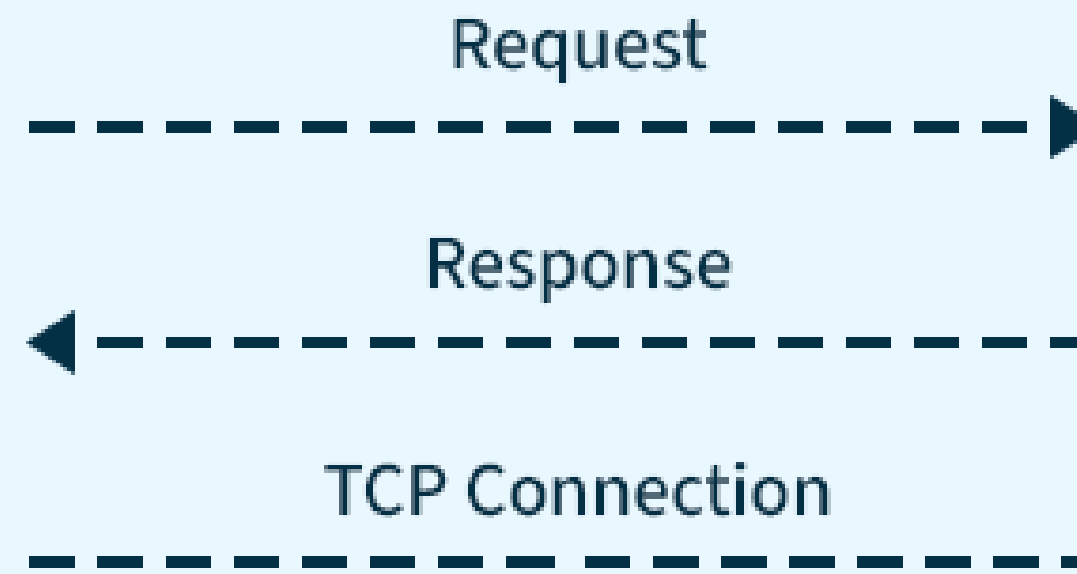
Used to establish a tunnel to the server through an HTTP proxy, commonly used for SSL/TLS connections.

HTTP Request/Response:

HTTP Connection



Client



Server

Hypertext Transfer Protocol - Secure (HTTPS)

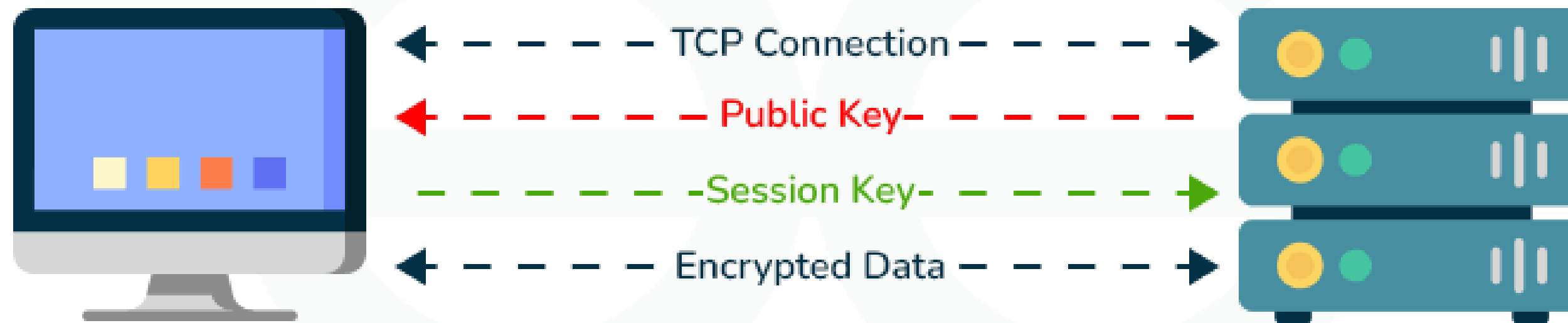


HTTPS stands for HyperText Transfer Protocol Secure. It is the most common protocol for sending data between a web browser and a website. HTTPS is the secure variant of HTTP and is used to communicate between the user's browser and the website, ensuring that data transfer is encrypted for added security.

Any website, especially those requiring login details, should use HTTPS. You can see a padlock icon in the URL bar, which means the page is secure. Browsers, like Google Chrome, treat HTTPS seriously and mark non-HTTPS websites as "Not Secure."

How Does HTTPS Work?

HTTPS



Use Case :- *Web Browsing*



Why HTTPS Matters and What Happens Without It?



HTTPS is important because it keeps the information on websites safe from being easily viewed or stolen by anyone who might be spying on the network. When a website uses regular HTTP, data is sent in small chunks called packets that can easily be intercepted using free software. This makes communication, especially over public Wi-Fi, very vulnerable to attacks.

On the other hand, HTTPS encrypts the data, so even if someone manages to intercept the packets, they will appear as random, unreadable characters. For example:

- Before encryption:

```
"This is a string of text that is completely readable"
```

- After encryption:

```
"ITM0IRyiEhVpa6VnKyExMiEgNveroyWBP1gGyfkf1YjDaaFf/Kn3bo3OfghBPDWo6AfSH1NtL8N7ITEwIXc1gU5X73x  
MsJormzzXlw0yrCs+9XCPk63Y+z0="
```


What is a Web Browser?



A Web Browser is a software application that allows users to view and interact with websites on the Internet. It retrieves data from a web server, processes it, and displays it in a readable and interactive format for the user.

Features of a Web Browser:

- **HTML Rendering:** Renders web pages written in HTML, CSS, and JavaScript.
- **Multimedia Support:** Supports images, videos, and other multimedia content.
- **Tabs:** Allows multiple websites to be opened in different tabs.
- **Extensions:** Supports browser extensions for additional functionality (e.g., ad blockers, password managers).
- **Security:** Includes features like private browsing, HTTPS encryption, and security certificates.



What is a Web Server?



A Web Server is a computer system that stores, processes, and delivers web content to users through the Internet. It listens for requests from clients (web browsers), processes those requests, and sends the required files (such as HTML, images, and videos) back to the browser.

Features of a Web Server

- **HTTP/HTTPS Protocol:** Communicates using HTTP or HTTPS to send and receive requests.
- **Content Storage:** Stores web pages, scripts, images, and other media.
- **Response Handling:** Processes incoming requests and returns the appropriate response (such as HTML pages or images).
- **Logging:** Logs details of every request, such as IP addresses, timestamps, and requested URLs.
- **Security:** Handles security measures like SSL/TLS encryption for secure HTTPS connections.